BinSight: A Lightweight, Vision-based Waste Sorting System

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1 Introduction

Waste generation is one of the largest contributors to climate change and carbon emissions. Every year, the world generates over 2 billion tons of waste [6]. This waste generates over 1.5 billion metric tons of CO_2 equivalent greenhouse gases [6]. Incorrect waste disposal practices play a major role in this issue; in the U.S., the average recycling contamination rate (number of non-recyclable items that are placed in the recycling) is 1 in 4 items, or 25% [7]. Although 75% of waste is recyclable, only 34% is actually recycled [17].

Errors due to confusing bin systems and lack of waste disposal knowledge are a major source of this problem. Despite major investments in waste disposal education, efforts have yet to demonstrate a significant improvement to this end—recent surveys find that over 61% of consumers in the U.S. have incorrect beliefs about proper waste disposal [11]. Signs and boards on waste bins can often be confusing and difficult to navigate, as well as easy to ignore.

Given these factors, new approaches to inform our waste disposal systems are crucially needed. Two notable efforts have been made in recent research. One approach is the smart-bin system [12], [18], [24], which involves a large, commercial or industrial-grade automated binning system, requiring large-scale manufacturing and costing thousands of dollars. The high-cost and major hardware requirement has deterred the wide-spread adoption of these systems. Another approach is the use of mobile phone applications to classify waste prior to disposal [8]. While this solution is cheap and widely available, it requires the user to have the time and sensitivity to take out their phone and classify the waste in their hand. Given that waste disposal is typically an on-the-go, reflexive task for most people, this poses a barrier to the actual rate of usage and practicality of the system.

Given these constraints, we propose a novel, lightweight waste-sorting solution using inexpensive hardware and vision-based inference. We train and deploy a convolutional neural network on a full-stack pipeline to produce low-latency waste bin predictions, all running on a minimal hardware setup. Through user feedback and human-centered design considerations, we curate our waste disposal messages to be fault-tolerant and user-friendly manner, accounting for prediction uncertainty.

2 Problem Statement

Our goal is to increase the rate of proper waste disposal and confer waste disposal knowledge to the general population using a lightweight, publicly deployable solution. AI, with its real-time visual recognition capabilities, can instantly classify waste items and guide individuals towards proper disposal, encouraging informed recycling decisions without human intervention. We contribute a low-cost AI solution to convey waste disposal information in a concise, impactful, and user-friendly manner.

3 Interface Design

3.1 User Interface

We developed the user interface using Next.js with React for frontend development and Chakra UI for styling. The website is fully operational and can be accessed here.

3.1.1 Design Philosophy

Our design philosophy centered around simplicity and intuitiveness. The interface needed to be straightforward enough for any user, ensuring a smooth and efficient user experience. This was particularly important given the reflexive nature of waste disposal, where users are likely to interact with the system for only a brief period.

3.1.2 User Experience Flow

We intend for the user experience to have a seamless flow, minimizing effort and maximizing efficiency. The process unfolds as follows:

1. **Activation and Capture:** Upon activation, a capture screen is displayed with the webcam feed. A bounding box prompts users to place their waste item within it. The system is programmed to detect stability in the bounding box area, ensuring accurate image capture. We tuned bounding box sizes to find a region that worked best across various object depths and webcam resolutions.
2. **Image Processing:** Once the item is stable in the bounding box for the set duration, the image is captured and sent to the backend for classification. We carefully tuned the stability threshold to balance latency (allowing for natural movement or background noise) with image quality, ensuring the object is mostly stable before capturing.
3. **Loading Screen:** During classification, a loading screen is presented, indicating the processing of the item.
4. **Classification Display:** Depending on the success of the classification process, the system either displays the classification result or an error message. A successful classification leads to a screen displaying the type of waste and its disposal method.
5. **Classification Feedback:** We incorporate conditional rendering based on the predicted class and its confidence score. The system categorizes confidence levels into three: low (below 60%), medium (60% to 80%), and high (above 80%). Corresponding messages are displayed alongside the classification to guide users further, as shown in the figure below. For instance, a low-confidence trash classification would display "MAYBE: TRASH" along with a message suggesting adjustments for clearer view.

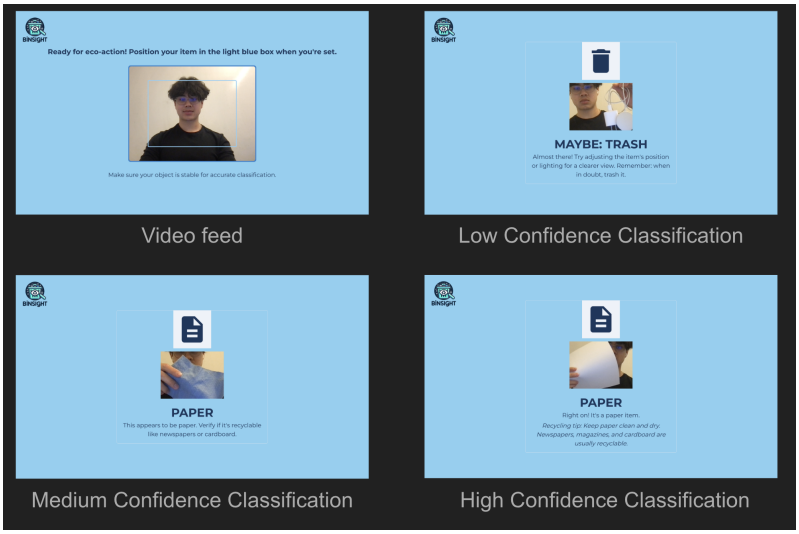


Figure 1: Examples of the UI and classification displays based on confidence.

6. **Reset to Capture Screen:** After displaying the classification for a short duration, allowing the user to dispose of their item correctly, the system reverts to the capture screen, ready for the next user. We also tuned the message display time to find a natural balance between information intake and user patience.

3.1.3 Backend Integration

The interface communicates with a server-hosted machine learning model via a Flask API. This setup enables efficient processing and classification of the waste item images captured by the frontend, ensuring a smooth and uninterrupted user experience.

3.2 Challenges

A major implementation challenge we faced was developing the frame selection process: implementing the frame-wise stability metrics and properly displaying and extracting from an ROI within the video feed, as well as tuning the stability hyperparameters (sampling rate, stability threshold, required stability time) for a suitable user experience. Additionally, we initially had trouble setting up the server API to ensure we could properly communicate with and make calls to the backend without running into network blockages. The interface underwent several UI modifications following our user study; further details about the evolution of our UI along with modifications we made is documented in section 6.2.3 below.

3.3 Hardware

The selection of hardware components is vital for ensuring the portability, cost-effectiveness, and seamless integration of our user interface into diverse settings, such as university learning commons. Our choice of hardware components was motivated with the goal of achieving a balance between performance, cost, and ease of integration. Below is a detailed overview of each component and its role in the system.

1. **Raspberry Pi 4 Model B - 4GB:** The heart of our system, the Raspberry Pi 4 Model B with 4GB of RAM, offers a balance of computational power and affordability. Its quad-core processor and sufficient RAM are ideal for running the website in browser and displaying it on a monitor for our waste classification application, handling real-time data processing and camera configuration.

2. **MicroSD Card:** We chose the Samsung 32GB PRO Endurance MicroSDHC for its storage capacity and fast read/write speeds essential for efficient operating system loading and data processing. The microSD card serves as the primary storage for the operating system and the application data on the Raspberry Pi.

3. **HDMI Adapter:** The Monoprice 6 in. 34AWG High Speed HDMI Cable with Ethernet Port Saver was selected to connect our system to a monitor. This adapter is crucial for setting up the Raspberry Pi as well as for direct interaction with the system and the monitor.

4. **Case & Power Adapter:** The Micro Connectors Aluminum Case Kit with Power Adapter and Fan is critical for protecting the Raspberry Pi from physical damage and ensuring its longevity. The included fan is essential for maintaining optimal operating temperatures, thus preventing thermal throttling and ensuring consistent performance.

5. **Camera:** The Raspberry Pi Camera 3 is used for capturing images of waste items. Its compatibility with the Raspberry Pi and its HDR and 12MP resolution make it ideal for our application, where accurate image capture is paramount for effective waste classification.

6. **Motion Sensor:** We experimented with incorporating the Stemedu HC-SR501 PIR Infrared Motion Sensor to activate the system upon detecting motion. Although we were not able to find a suitable deployment paradigm for this motion sensor, this component is beneficial for energy efficiency and privacy, as it allows the system to be on and capturing only when activation is needed.

In addition to the previously mentioned components, our project required a monitor, HDMI cable, keyboard, and mouse for initial setup, debugging, and direct interaction with the system. For the purposes of our project, these items were readily available from team members, thus incurring no additional cost. However, it's important to note that any standard monitor, HDMI cable, keyboard, and mouse would be suitable for this project.

1. **Monitor:** In scenarios where a monitor is not readily available, a suitable choice would be the Kuman 7 Inch IPS Display 1024x600 LCD Screen Computer Monitor. This monitor is compatible with various Raspberry Pi models and offers a high resolution of 1024x600 pixels.

2. **HDMI Cable, Keyboard, and Mouse:** Any standard HDMI cable, keyboard, and mouse would suffice for this system. It should be noted that the keyboard and mouse would only be required for booting and setting up the Pi, and would not be required for sustained operation of the Pi, which would be deployed headless.

Component	**Cost**
MicroSD Card	$6.99
HDMI Adapter	$10.99
Case & Power Adapter	$20.99
Raspberry Pi 4 Model B	$54.99
Camera	$24.99
Motion Sensor	$10.06
Kuman 7 Inch Monitor	$37.99
Total	**$166.00**

Table 1: BinSight Hardware Components Cost Breakdown

The total cost, $166.00, remains highly economical, considering the functionality and versatility of the system, and compared to traditional hardware-based waste sorting systems [12], [18], [24].

4 Machine Learning Backend

4.1 ML Design: Initial Proposal

This section describes the details of our initial proposal for AI garbage detection and classification module (See Fig. 2), including pose estimation (Sec. 4.1.1), cropping (Sec. 4.1.2), garbage classification (Sec. 4.1.3), and the infrared motion sensor (Sec. 4.1.4).

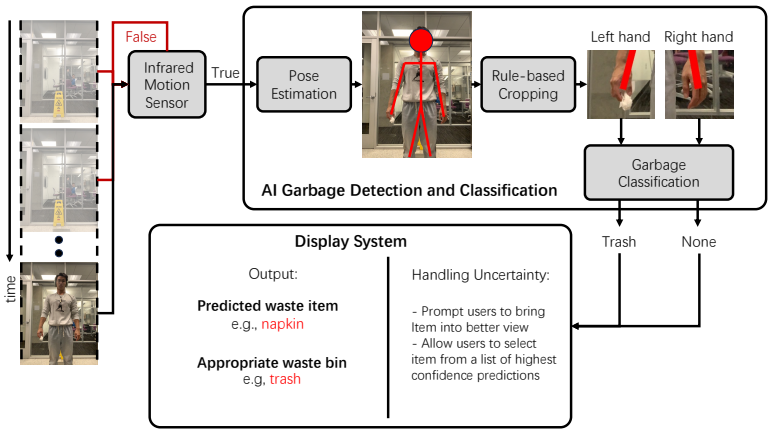


Figure 2: The pipeline of the initial proposed method.

4.1.1 Pose Estimation

Current garbage detection works [4, 21] try to locate the garbage from an underwater or an unmanned aerial vehicle's view. However, the complex background, occlusion, and human interruption in our setting make the direct usage of the above methods unfeasible. As a result, in contrast to current works that directly detect/segment the garbage in the image, we aimed to conduct pose estimation to locate the right-/left-hand as potential areas of the trash. To achieve this, we utilized pre-trained pose estimation methods [23].

4.1.2 Cropping

With the right-/left-hand detected, we further aimed to crop the corresponding bounding boxes as the potential area of the trash since it is most likely that the user is holding the garbage. Furthermore, it can enhance the efficiency and effectiveness of garbage classification since we eliminate the interruption and decrease the size of the input to the garbage classification model. We designed a cropping strategy that is flexible for edge cases and variation in trash size and form, as well as account for occluded body parts.

4.1.3 Garbage Classification

Having the trash area cropped, we send it to the garbage classification model to get the waste sorting results. However, current garbage classification methods [19, 9] focus on classifying the trash in a relatively clear background, such as images taken in the waste sorting machine, which are not necessarily representative of the ones we are dealing with. Consequently, we trained and fine-tuned a Convolutional Neural Network such as EfficientNet [16] with mixed datasets with data augmentation and added noise to enhance the robustness of the model.

4.1.4 Infrared Motion Sensor

Regarding the system interacting with the users, we planned to use an infrared motion sensor to activate the request to the AI model. This would reduce the delay resulting from frequent requests, prevent privacy issues from constant video monitoring, and save substantial power from camera operation. With the lower computational costs, we would be able to reduce the cost and more widely deploy the proposed system.

4.2 ML Design: Updated Version

Our initial design proposed a pose estimation (Sec. 4.1.1 and Sec. 4.1.2) system to identify a section of the image containing a waste item for which to classify. We modified this to use an object detection model instead (See Sec. 4.2.1), as pose estimation would over-complicate the frame selection process, and object detection allows for more flexibility and potentially more robustness.

We also opted to include a bounding box on the video stream in which the user could display their item, allowing for an interactive design element that would reduce the noisiness of the selected frames, while still requiring relatively little effort on the part of the user.

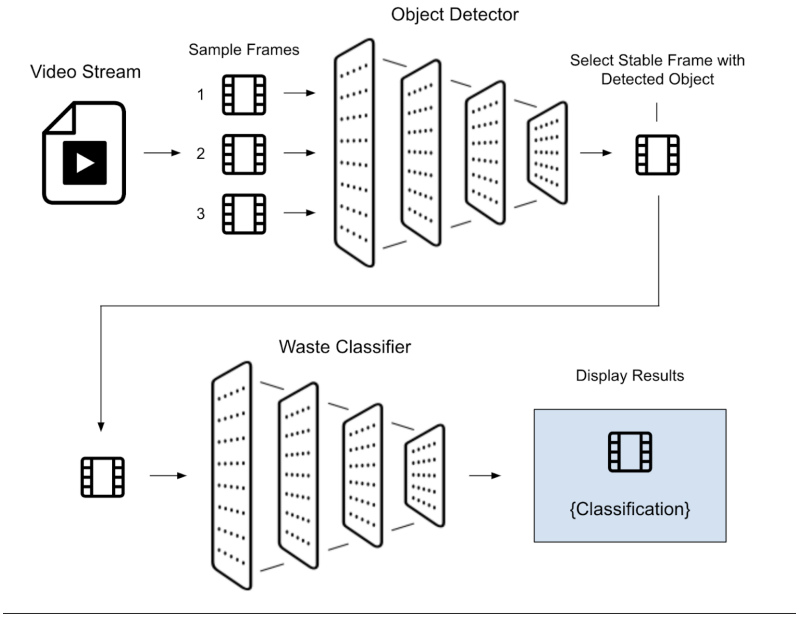


Figure 3: The pipeline of the updated method.

4.2.1 Trash Detector

As shown in Fig. 3, we used Yolo-v8 as the detector to detect the trash object within the image and adopted EfficientNet-B2 [16] to classify the correct bin the trash belongs to. Specifically, we first used the detector to sample the frame once the predicted bounding box is stable for N frames then we cropped the image according to the bounding box and sent it to the classifier. Since we currently adopt the COCO [10] dataset pre-trained weights to conduct the inference, we simply aggregate certain classes, *e.g.* bottle, wine glass, cup, *etc*, in the COCO dataset to identify the trash as the target objects.

4.3 ML Design: Final Version

We transitioned to a pixel-wise stability metric rather than using Yolo to minimize latency, avoid erroneous object detections, and simplify the overall pipeline. Our final pipeline is shown in Fig. 4. We use a pixel-wise difference across frames as the sampling strategy, and adopt EfficientNet-B2 [16] to classify the correct bin the trash belongs to.

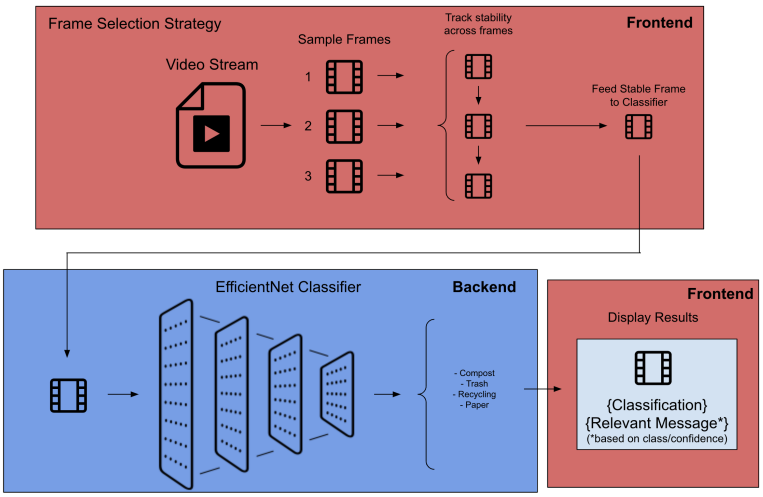


Figure 4: The pipeline of the final proposed system.

4.3.1 Pixel-wise Stability Metric

Specifically, we sample the frame every 500ms. Once the image is stable (within a pixel-wise difference of 25 across the image) for 1500ms, we crop the image according to the bounding box on the video feed and send it to the classifier.

4.3.2 EfficientNet Classifier

Next, we use EfficientNet-B2 to classify the correct bin *i.e.* Trash, Recycle, Compost, and Paper. EfficientNet is obtained through neural architecture search based on MnasNet [15], where a.) the convolution with a stride of 2 is adopted for downsampling instead of maxpooling at the beginning, b.) the channel after the first downsampling is reduced, c.) many stages use a 5×5 convolution, and d.) the size of the feature map after downsampling is halved, but the channel is not doubled. The above modifications not only reduce information loss but are also less computationally intensive, which is better than ResNet [5] or MobileNetV2 [13].

5 Data

The imbalanced data distribution from TACO dataset [1] that we used in the updated version of our ML design caused poor performance and biased prediction in classifying trash Fig. 5.

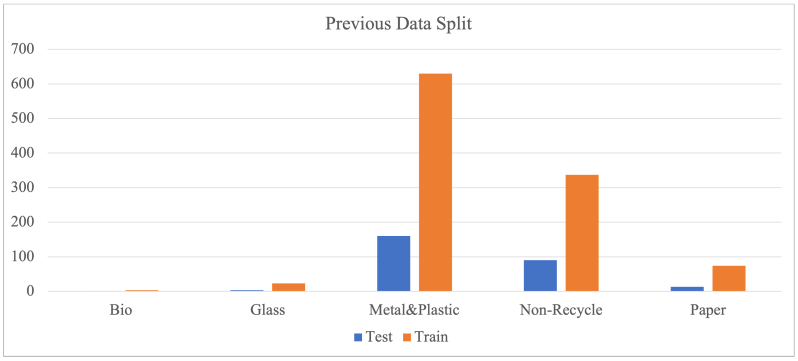


Figure 5: Previous Data split from TACO dataset [1]

As a result, we seek to curate a larger dataset to train the model and, more importantly, fit the JHU's recycling rules in the final version of our ML design. To this end, we first adopt a composite dataset from waste-sorting, which includes MJU-Waste [20], TrashNet [22], CompostNet [3], and Fruits and Vegetables Image Recognition [14]. MJU-Waste [20] is the largest public benchmark available for waste object segmentation, which is created by collecting waste items from a university campus, bringing them back to a lab, and then taking pictures of people holding waste items in their hands. All images in the dataset are captured using a Microsoft Kinect RGBD camera. TrashNet [22] is the first dataset pertaining to garbage materials, where the data acquisition process involved using a white poster board as a background and using sunlight and/or room lighting. The devices used were Apple iPhone 7 Plus, Apple iPhone 5S, and Apple iPhone SE. CompostNet [3] added photos of food waste and landfill waste based on TrashNet. The photos of the waste were taken against a white poster board using natural or overhead lights and focused on one piece of waste in each photo. Last, Fruits and Vegetables Image Recognition [14] contains images of fruits and vegetables. The images in this dataset were scraped from Bing Image Search.

To sum up, the waste-sorting dataset contains 1,797, 1,929, and 1,012 samples for Compost, Recycle, and Trash, respectively. However, to fit JHU's recycling rule, we manually split and reassemble the data one by one and form our dataset (Compost: 1,982, Trash: 1,570, Recycle: 590, Paper: 571) as shown in Fig. 6. Additionally, we adopt data augmentation of RandomCrop, HorizontalFlip, VerticalFlip, ShiftScaleRotate, RandomBrightnessContrast, and Cutout from Albumentations [2] during the training to increase the robustness.

Even though our composite dataset contains various distributions from different datasets, there are some advantages and disadvantages that can be discussed and improved. For example, the model's robustness can be enhanced by leveraging the diversity within our composite dataset. Our dataset includes both naturalistic images (displaying a person holding trash in a naturalistic background) alongside web-scraped images (clear clip-art images of fruit); dealing with these inconsistencies could help improve the robustness of our system. Additionally, new waste classes, expanding from JHU's recycling rules, would be needed for flexibility of deployment in new settings. The bias in the number of images in different classes is also a potential limitation (See Table 2). In future work, we could collect waste items from JHU's campus and curate our own data, producing consistent naturalistic images that fit the deployment setting to mitigate class biases and enhance the robustness of our model.

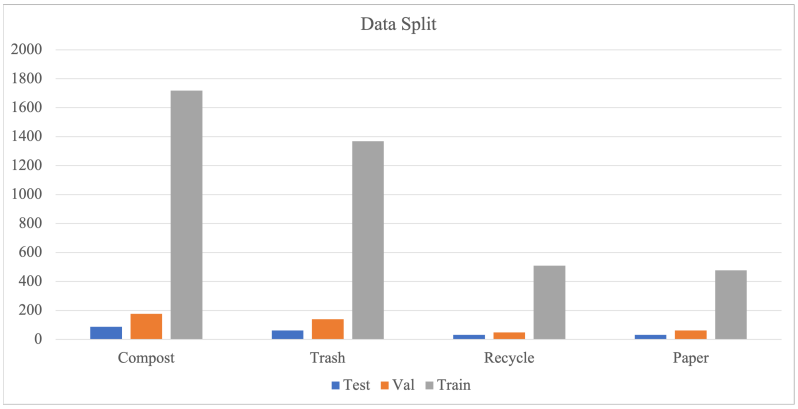


Figure 6: Reassembled Data split from Waste-sorting

6 Evaluation

Evaluating the effectiveness and efficiency of BinSight involves a multifaceted approach, where we must consider both the performance of the machine learning backend and the overall system functionality. This dual-mode evaluation is crucial to ensure not only the accuracy and reliability of our AI-driven waste classification but also the practicality and user-friendliness of the system as a whole. We aim to provide a comprehensive assessment of BinSight's performance, which guided certain decisions in the design process as well as highlighting areas of success and identifying potential improvements.

6.1 Machine Learning Backend Evaluation

We evaluate the classification model on the test set and report the model performance for each class separately for a better analysis. To provide a deeper insight into the model performance, we use the following nine different metrics for model evaluation: (a) Accuracy: Represents the overall correctness of predictions. (b) F1 Score: Balances precision and recall, suitable for imbalanced datasets. (c) True Positive Rate: Measures the proportion of actual positives correctly identified. (d) True Negative Rate: Measures the proportion of actual negatives correctly identified. (e) Positive Predictive Value: Indicates the accuracy of positive predictions. (f) Negative Predictive Value: Indicates the accuracy of negative predictions. (g) False Positive Rate: Measures the proportion of actual negatives incorrectly classified as positives. (h) False Negative Rate: Measures the proportion of actual positives incorrectly classified as negatives. (i) False Discovery Rate: Represents the proportion of positive predictions that are incorrect.

As shown in Table 2, our model achieves satisfactory performance on all classes. However, the result in the paper and recycle class is generally lower in F1 Score and Sensitivity. This suggests our classification model may be prone to missing actual positive instances (decreased Sensitivity) while struggling to achieve accurate positive predictions and prevent false positives (decreased F1 Score), indicating challenges in handling positive instances. This can also be seen from Fig. 7, where the prediction for the paper and recycle class is prone to the error class: trash. We believe that the bias prediction results from the imbalance data distribution as mentioned in Sec. 5, which can be improved in future work.

Metrics	Compost	Paper	Recycle	Trash	Avg.
(ACC) Accuracy	**0.948**	0.939	0.943	0.877	0.927
$(F1)$ F1 Score	**0.936**	0.772	0.769	0.819	0.824
(TPR) True positive rate (Sensitivity/Recall)	0.920	0.688	0.645	**0.952**	0.801
(TNR) True negative rate (Specificity)	0.968	0.983	**0.994**	0.847	0.948
(PPV) Positive predictive value (Precision)	**0.952**	0.880	**0.952**	0.720	0.876
(NPV) Negative predictive value	0.945	0.947	0.942	**0.977**	0.953
(FPR) False positive rate (Fall out)	0.032	0.017	**0.006**	0.153	0.052
(FNR) False negative rate	0.080	0.313	0.355	**0.048**	0.199
(FDR) False discovery rate	**0.048**	0.120	**0.048**	0.280	0.124

Table 2: Quantitative results

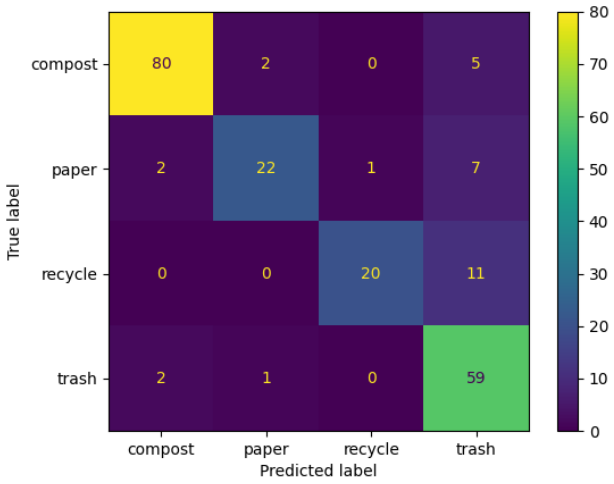


Figure 7: Confusion Matrix

6.2 System Evaluation

We evaluated the system based on the effectiveness, usability, and practicality of the system in real-world scenarios. Some key aspects that we considered during this evaluation were latency of classification and user feedback as a whole. We conducted experiments carefully designed to address such aspects, followed by an analysis of the results.

6.2.1 System Evaluation: Experimental Setup

In order to collect the data we need, we conducted a user study, targeting how well the full process was integrated, and identifying pain points in the user experience. It consisted of setting the hardware up in its entirety in the Atrium of Brody Learning Commons, along with the classic four bins for waste disposal that are present around Johns Hopkins campus, and inviting students in the area to dispose of their waste with our system. We also invited students to use the traditional bins without our system, and asked about their experience to serve as a baseline. We gave the following survey of questions to each of the users, where 5 is the best rating and 1 is the worst rating:

No.	Survey Question
1	How easy was it to understand the use of the system? (1-5)
2	How helpful was this system in informing your waste disposal decision? (1-5)
3	How confident were you in your waste disposal decision? (1-5)
4	How quickly did you make an informed waste disposal decision? (1-5)
5	What did you like most about the system and what needs improvement?

Table 3: User Feedback Survey Questions

We queried 15 students around Brody Learning Commons and compiled their results, comparing the responses with the traditional bins and with the BinSight system.

6.2.2 System Evaluation: Quantitative Results

The results of these experiments are shown below. We will discuss our takeaways and subsequent modifications based on these results in our analysis below.

No.	Survey Question	BinSight	Traditional
1	How easy was it to understand the use of the system? (1-5)	3.60	4.47
2	How helpful was this system in informing your waste disposal decision? (1-5)	3.73	2.33
3	How confident were you in your waste disposal decision? (1-5)	3.67	3.01
4	How quickly did you make an informed waste disposal decision? (1-5)	2.33	2.53

Table 4: User Responses

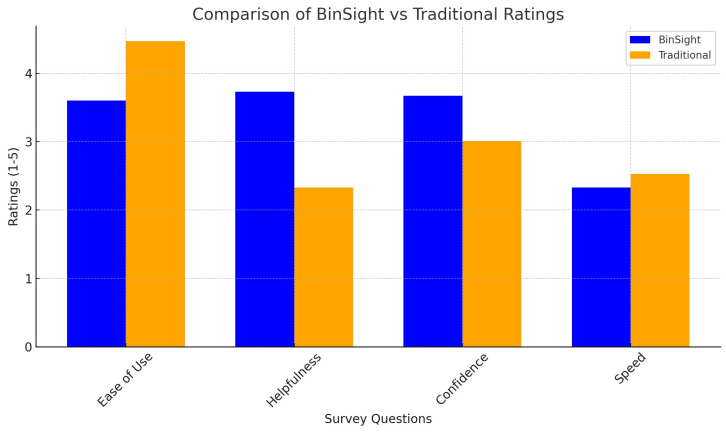


Figure 8: User Response Histogram

6.2.3 System Evaluation: Qualitative Results

The open-response questions provided a means for users to provide a qualitative evaluation of the system and give meaningful insight to how we can further improve BinSight. Summarizing the key aspects of the responses, in tandem with the quantitative results above, we can draw the following conclusions:

1. **Traditional bin systems are straightforward, but flawed.** Although traditional bins are straightforward to use, we confirmed our initial suspicions that users are unsatisfied with traditional bin labels. Users said that traditional bin labels are generally not informative enough, and only show a small sample of waste examples, making it hard to generalize to new waste items.

2. **BinSight is more informative.** Users on average felt that BinSight provided more information compared to the traditional method, and felt more confident about their waste disposal decisions with our system.

3. **BinSight latency was a problem.** The biggest gripes with either of the systems was when something took too long. Slow processing frustrated users, often because the stability requirement was too strict (users would get impatient or fidgety), or due to hardware or network buffering.

4. **BinSight instructions need clarity.** Occasionally, users misunderstood the stability requirement entirely due to lack of instruction clarity, and would wave the item around in front of the camera, never reaching a classification.

These results prompted us to make the following modifications to our system to better align with the desired user experience.

1. **Instruction clarity:** We included an instruction guiding the user to hold the object stable in our video feed UI to ensure users understood the stability requirement.

2. **Latency improvements:** We tweaked the stability threshold to properly balance frame quality with natural movement of the object. Additionally, we tweaked the time the message is displayed to find a natural balance between information intake duration and user patience.

3. **Classified image display:** Users suggested showing the image that was classified along with the message to provide classification interpretability.

4. **Adding uncertainty for low-confidence predictions:** We noticed in some cases that users would not read the message and only looked at the primary bin label message to make their decision. To ensure users were aware of the uncertainty in low-confidence cases, we added a "Maybe:" prior to our predicted bin in the case of low-confidence predictions, preventing blind adherence to system suggestions.

These modifications aimed to better align our system with the desired experience of our users, while addressing their frustrations - improving the latency, UI design, and ease of use of our system.

7 Implications

This section outlines the realized implications of BinSight, focusing on the aspects we successfully implemented and the impact these have on various stakeholders, including users and entities deploying the system.

7.1 Lightweight System

We successfully implemented a lightweight model for waste classification, meeting our initial goals of energy efficiency and cost-effectiveness. The deployment of the Raspberry Pi 4 Model B as the core hardware component ensured low power consumption and minimal computational resources. This aligns with our aim of reducing both economic and environmental costs, making the system ideal for wide adoption.

7.2 User Experience and Stakeholder Engagement

We designed our interface with simplicity and intuitiveness in mind. Features such as feedback based on classification confidence scores directly engage users, making waste disposal an informative and interactive process. This aligns with our goal of promoting effective waste management and recycling.

Some drawbacks of the user experience include the aforementioned latency issues, caused by both the frame selection method using stability. Future deployments could make use of object detection or segmentation models for more immediate and adaptive frame selection, improving both the latency and frame quality of the extracted image. However, these methods may beget hardware with greater processing power to handle the inference demands.

We preserve user privacy by never saving user images beyond a classification sequence. Although we experimented with using motion-sensors and ultimately failed to find an adequate deployment paradigm, motion-sensing would allow the system to only be activated when prompted by close contact, not only saving power, but also avoiding constant video monitoring in public areas.

While we proposed gamification elements like a point-based system for encouraging recycling behaviors, these were not implemented in the current version of BinSight. Future iterations could explore these aspects to further engage users.

7.3 System Flexibility and Practical Deployment

The system's flexibility for different deployment scenarios has been a crucial achievement. Our choice of hardware and software components allows for easy adaptation to various settings, such as educational institutions and public spaces. The Raspberry Pi's portability and the modularity of our software design ensure that deployment of the BinSight system is straightforward and easily adoptable.

However, a potential drawback include performance variations in diverse environments. As discussed in the evaluation of our ML backend, our model could be improved with greater dataset diversity, accounting for variations in background and object depth. If we wish to deploy the system in other areas of the nation or world, further training would be required to ensure the classifier properly generalizes to trash items from different cultures.

Additionally, our hardware and monitor system would require weather and damage proofing if it were to be deployed outdoors on public wastebins. Case-protected iPads could be a suitable display format.

8 Timeline and Summary of Milestones

1. Select and clone selected model repos, load and analyze selected datasets into code.
2. Initial model training/fine-tuning on datasets, initial classification performance evaluations.
3. Begin setting up the front-end website and develop the frame selection implementation.
4. Improve model performance, begin integration of front-end and model infrastructure for overall system.
5. Perform more detailed model evaluations, work on testing the integrated software components and finalize the system infrastructure.
6. Finalize overall system design, perform internal UI evaluations, identify sources of error and critical areas of improvement.
7. Integrate the full system pipeline onto the Raspberry Pi and relevant hardware, including the display and PiCam.
8. Refine the frontend display for adaptive messages based on confidence and helpful tips. Setup the localhost to run on the Raspberry Pi.

9. Test and ensure that the backend API functions with Raspberry Pi requests. If time permits, continue refining the classifier.
10. Evaluate the system in real-world scenarios, deployed on waste bins at Hopkins, recruiting participants and evaluating the usability and effectiveness of the system.
11. Present final results.

As we prioritized fully integrating our system over refining smaller technical details, some of the above milestones still leave room for future improvement: for instance, refining the classifier with an enhanced dataset for system robustness, producing an ML based frame selection method for improved frame quality, and implementing additional hardware components (such as motion sensing). Ultimately, though our methodologies changed over the course of the project, we were able to successfully cover all of our projected milestones: developing the ML solution and full-stack application, deploying on hardware, and conducting a thorough user evaluation. We hope that our project represents a step forward in AI-driven solutions dedicated to enhancing environmental health and sustainability.

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